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Structural foundation for the role of enterococcal PrgB in conjugation, biofilm formation and virulence

Wei-Sheng Sun, Lena Lassinantti, Michael Järvå, Andreas Schmitt, Josy ter Beek, Ronnie P-A Berntsson

Department of Medical Biochemistry and Biophysics, Umeå University, SE-90187 Umeå, Sweden • Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden

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Abstract

Type 4 Secretion Systems are a main driver for the spread of antibiotic resistance genes and virulence factors in bacteria. In Gram-positives, these secretion systems often rely on surface adhesins to enhance cellular aggregation and mating pair formation. One of the best studied adhesins is PrgB from the conjugative plasmid pCF10 of *Enterococcus faecalis*, which has been shown to play major roles in conjugation, biofilm formation and importantly also in bacterial virulence. Since *prgB* orthologs exist on a large number of conjugative plasmids in various different species, this makes PrgB a model protein for this widespread virulence factor. Here we report structures for almost the entire PrgB, in the presence or absence of DNA, using a combination of X-ray crystallography and cryo-EM. These reveal that PrgB undergoes a large conformational change upon DNA-binding and that it contains four immunoglobulin-like domains. We re-evaluate previously studied variants and present new *in vivo* data where specific domains or conserved residues have been mutated. For the first time we can show a decoupling of cellular aggregation from biofilm formation and conjugation in *prgB* mutant phenotypes. Based on the presented data, we propose a new functional model to explain how PrgB mediates its different functions. We hypothesize that the Ig-like domains act as a rigid stalk that both protect the previously studied polymer adhesin domain from proteolysis, as well as presenting it at the right distance from the cell wall.

eLife assessment

This study presents a **valuable** combination of X-ray and cryo-EM structures of the bacterial adhesin PrgB, an atypical microbial cell surface-anchored polypeptide that binds DNA. There is **convincing** support for the claims regarding the overall function and importance of individual domains. The model for PrgB's binding of eDNA is thought-provoking, but the evidence for it based on low-resolution volumes of cryoEM data is **incomplete**. If additional experimental evidence for the model is produced, this work will be impactful in the field of bacterial adhesins, conjugation, and biofilm formation, and is particularly **important** as it focuses on a clinically **important** Gram-positive pathogen, whereas most work in the field has been focused on Gram-negative model systems.

Introduction

Enterococcus faecalis (*E. faecalis*) is one of the leading causes of hospital acquired infections, such as urinary tract infections and endocarditis^{(1),(2)}. These infections are difficult to treat as *E. faecalis* has the tendency to form biofilms and is often resistant to various antibiotics. They are also notorious for spreading antibiotic resistance and other fitness advantages by transfer of mobile genetic elements (MGEs), which can be located on conjugative plasmids or in the chromosome^{(3),(4)}. Conjugative plasmids usually also encode a Type 4 Secretion System (T4SS) that mediates its transfer, via conjugation, from a donor cell to a recipient cell^{(5)–(8)}. However, conjugative plasmids and their T4SS have almost exclusively been studied in Gram-negative model systems⁽⁸⁾.

One of the few well-characterized Gram-positive conjugative plasmids is pCF10 from *E. faecalis*^{(6),(9),(10)}. This conjugative plasmid contains a ~27 kbp operon that is tightly regulated by the P_Q promoter^{(11)–(14)} and that encodes all proteins needed for conjugation. This operon also encodes three cell-wall anchored proteins: PrgA, PrgB, and PrgC. PrgA is a conjugation regulator that provides surface exclusion to prevent unwanted conjugation. We have previously shown that PrgA consists of a protease domain that is presented far away from the cell wall via a long stalk and that it's likely mediating the proteolytic cleavage of PrgB^{(15),(16)}. PrgC is a virulence factor, but its function and structure remain unknown⁽¹⁷⁾. PrgB is the main adhesin produced by pCF10 and has been studied for well over 3 decades. This protein is around a 140 kDa in size and possesses an N-terminal signal sequence and a C-terminal LPXTG cell wall anchor motif⁽¹⁸⁾. PrgB distributes over the entire surface of the cell-wall and increases cellular aggregation, biofilm formation and the efficiency of plasmid transfer^{(19),(20)}. Several mammalian infection model systems have shown that PrgB is a strong virulence factor^{(18),(21)–(26)}. One reason for this virulence is that PrgB mediates biofilm formation in an extracellular DNA (eDNA) dependent manner⁽¹⁷⁾. Homologs of PrgB have been identified in many other conjugative plasmids⁽¹⁶⁾, suggesting that PrgB-like proteins confer important roles in a large number of bacterial species^{(27)–(29)}.

PrgB was initially identified as one of the driving forces in cellular aggregation⁽¹⁹⁾. To understand how it mediated this process, previous research has tried to identify the various protein domains that are present in PrgB and evaluate their function(s). Two RGD (Arg-Gly-Asp) motifs were identified (see [figure 1A](#)), and found to be important for vegetation and biofilm formation in the host tissue environment^{(18),(23)}. The N-terminal half of PrgB was

of PrgB246-558 and showed that it has a lectin-like fold that was most similar to adhesins from various oral *Streptococci*. As these adhesins are known to bind various polymers, we subsequently referred to PrgB246-558 as the polymer adhesin domain⁽¹⁶⁾. We have shown that this domain can bind both LTA and eDNA in a competitive manner. Bound eDNA is thereby strongly compacted as it is wrapped around the domains positively charged surface⁽³²⁾. We therefore proposed that PrgB could use eDNA to promote cell-to-cell contacts, as an alternative to direct binding to the LTA from a recipient cell (Fig. 1B). As all described polymer adhesin domains, PrgB has a central ridge with a conserved cation binding site⁽¹⁶⁾,⁽³³⁾. In the homologous GbpC, from *Streptococcus mutans*, this site has been suggested to bind glucans⁽³⁴⁾. However, no interaction with glucans has been observed for PrgB or any other homologs⁽³⁵⁾. Thus, the importance of this conserved motif remains an open question.

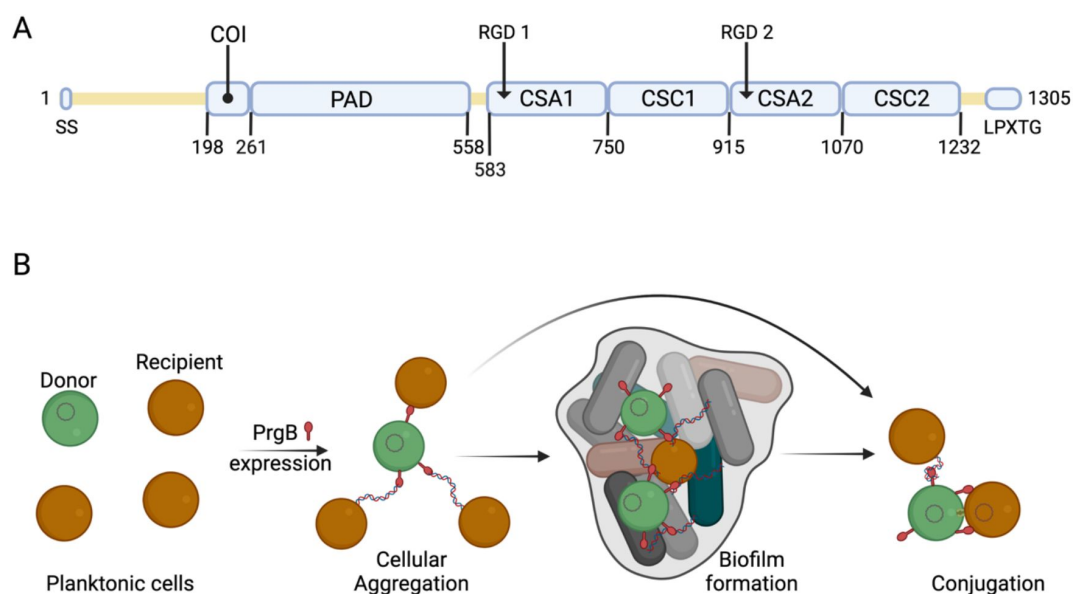


Figure 1.

Schematic overview of PrgB domain organization and function. A) Updated schematic overview of the domain organization of PrgB. SS: Signal sequence, COI: Coiled-coil, PAD: Polymer adhesin domain, CSA: adhesin isopeptide-forming adherence domain, CSC: cell-surface antigen C-terminal domain, LPXTG: cell wall anchor sequence. B) In a setting with a donor cell (green) and multiple recipient cells (brown), PrgB is produced and sits on the cell wall. There it enhances cellular aggregation and/or biofilm formation, either by directly binding LTA from the cell-wall of a recipient or first to eDNA to. PrgB compacts the eDNA, and thereby likely pulls the recipient cells closer. Once close enough, PrgB binds to the LTA of the recipient bacteria and facilitates mating-pair formation and conjugation.

The polymer adhesin domain plays an important role in the function of PrgB, but it only accounts for around a quarter of the entire protein. No structural data has been available for the remainder of PrgB. Here we present the structure of almost full-length PrgB, based on both X-ray crystallography and cryo-EM. This allows us to put the large amount of available phenotypic data into a structural context and explain a lot of previous observations. We also constructed several new mutants of *prgB* that better fitted the found domain organization and analyzed their *in vivo* effects on cellular aggregation, biofilm formation and conjugation efficiency. Based on our findings, we conclude with an updated model of how PrgB mediates its different functions.

Materials and Methods

Bacterial strains and growth conditions

See Table S1 for a full list of all strains, plasmids and oligonucleotides used. *Escherichia coli* Top10 was used in molecular cloning and grown in Lysogeny broth (LB). *E. coli* BL21 (DE3) was used for recombinant protein expression and grown in Terrific Broth (TB). The *E. faecalis* strains were cultured in Brain-Heart infusion broth (BHI) or Tryptic Soy broth without dextrose (TSB-D) as indicated in each assay. Concentrations of antibiotics for *E. coli* selection were as follows: ampicillin (100 µg/ml), kanamycin (50 µg/ml), spectinomycin (50 µg/ml), and erythromycin (150 µg/ml). In *E. faecalis* cultures, antibiotics were used as the following concentrations: tetracycline (10 µg/ml), fusidic acid (25 µg/ml), erythromycin (20 µg/ml for chromosome-encoded resistance; 100 µg/ml for plasmid-encoded resistance), spectinomycin (250 µg/ml for chromosome-encoded resistance; 1000 µg/ml for plasmid-encoded resistance), streptomycin (1000 µg/ml). Plasmids were transformed to *E. coli* by heat-shock transformation, whereas *E. faecalis* strains were transformed by electroporation⁽³⁶⁾.

Cloning and mutagenesis

To insert a multiple cloning site in pMSP3545S, DNA oligos of MCS_fwd and MCS_rev were resuspended in milliQ to 100 µM, mixed 1:1, denatured at 95°C for 15 min and slowly cooled to room temperature. The annealed MCS oligo was ligated into the gel-purified backbone fragment from pMSP3545S-*prgK* vector⁽³⁷⁾ digested with NcoI and XbaI (removing the *prgK* insert). This was done in a 30 µL ligation mixture with 90 ng of the digested vector and a 9 times molar excess of insert. 5 µL of this ligation mixture was transformed into Top10 competent cells and plated on LB agar plates with spectinomycin (50 µg/ml), and erythromycin (150 µg/ml). The constructed plasmid was analyzed by restriction digestion and sequenced to confirm the correct insertion of the multiple cloning site and is further called pMSP3545S-MCS.

pMSP3545S-*prgB* was constructed by PCR of *prgB* from pCF10 with the NcoI-*prgB*-F and BamHI-stop-*prgB*-R primer pair and placement into the pMSP3545S-MCS vector via NcoI/BamHI restriction, gel-purification of the correct DNA fragments and subsequent ligation. The constructed pMSP3545S-*prgB* was subsequently used as a template for mutagenesis creating pMSP3545S-*prgB* deletion or point mutation variants. This was carried out with inverse PCR (iPCR) using partially overlapping primer pairs. The iPCR products were gel-purified with the DNA clean-up kit and digested with DpnI to remove residual template plasmid DNA. The processed iPCR products were then transformed to Top10 competent cells. For overexpression, *prgB*₁₈₈₋₁₂₃₃ was cloned into the p7XC3GH vector using the FX cloning system⁽³⁸⁾. Mutations were introduced to p7XC3GH-*prgB*₁₈₈₋₁₂₃₃ with the same iPCR approach to obtain the derivative plasmid with *prgB*₁₈₈₋₁₂₃₃: S442A, N444A. All constructed plasmids were screened by PCR and verified by sequencing.

Protein purification and crystallization

PrgB was produced as previously described⁽³²⁾. Briefly, PrgB₁₈₈₋₁₂₃₃ was expressed with an N-terminal hexa-histidine tag from pET-*prgB*₁₈₈₋₁₂₃₃ or a C-terminal deca-histidine and GFP tag from p7XC3GH-*prgB* in *E. coli* BL21(DE3). The cells were grown at 37 °C in TB medium until they reached an OD₆₀₀ of 1.5. Then the temperature was lowered to 18 °C and protein production was induced by adding 0.5 mM IPTG. Cells were grown for 16 hours before harvesting by centrifugation. Cells were resuspended in 20 mM HEPES/NaOH (pH 7.0), 300 mM NaCl, 30 mM Imidazole and 0.02 mg/ml DNase I and broken with a Constant cell

disruptor at 4 °C and 25 kPsi (Constant Systems). The cell lysate was clarified by centrifugation for 30 minutes at 30,000 x g, 4°C and incubated at 4 °C with Ni-NTA (Macherey-Nagel). The Ni-NTA column was washed with 10 column volumes of 20 mM HEPES/NaOH (pH 7.0), 300 mM NaCl, 30-50 mM imidazole and bound proteins were eluted from the column with the same buffer supplemented with 500 mM Imidazole. The histidine affinity tags and the GFP when present, were cleaved off from the purified protein fractions by incubation with TEV protease (for pET-*prgB*) or Prescission protease (for p7XC3GH-*prgB*) in a 1:100 ratio for 20 h at 4 °C. The cleaved proteins were loaded on a Superdex-200 Increase 10/300 GL column (Cytiva) equilibrated in 20 mM HEPES/NaOH (pH 7.0) and 150 mM NaCl. The elution profile showed two peaks corresponding to a PrgB dimer and monomer⁽³²⁾. These two peak fractions were pooled separately and concentrated on an Amicon Ultra Centrifugal Filter with a 30 kDa cut-off (Merck-Millipore). 10% glycerol was added to the concentrated protein fractions, which were subsequently flash frozen in liquid nitrogen and stored at -80 °C.

Structure determination via X-ray crystallography

Purified PrgB₁₈₈₋₁₂₃₃ from the monomeric fraction, with a protein concentration of 15 mg/mL, were thawed and used in crystallization trials. Crystals of PrgB₁₈₈₋₁₂₃₃ were grown in 8-12 weeks, at 20 °C by sitting drop vapor diffusion in a condition with 0.2 M CaCl₂ and 20% PEG 3350 and a protein to reservoir ratio of 1:1 in the drop. Crystals were flash cooled in liquid nitrogen. X-ray diffraction data of PrgB₁₈₈₋₁₂₃₃ was collected on ID23-1 at the ESRF, France. The data was processed using XDS⁽³⁹⁾. The PrgB₁₈₈₋₁₂₃₄ crystals belong to the monoclinic space group P2₁2₁2₁ and contain two molecules in the asymmetric unit. The crystallographic phase problem was solved using molecular replacement using PHASER, using the SspB homology model of the Ig-domains as search models (PDB: 2WOY)⁽⁴⁰⁾. Further building of the model was conducted in COOT⁽⁴¹⁾. The structure was refined to 1.85 Å with crystallographic Rwork and Rfree values of 20.9/24.6 using Refmac5 and PHENIX refine⁽⁴²⁾,⁽⁴³⁾. The final PrgB₁₈₈₋₁₂₃₃ model consists of residues 584-1234, and was validated using MolProbity⁽⁴⁴⁾. Atomic coordinates and structure factors have been deposited in the Protein Data Bank (PDB Code: 8BEG).

Sample preparation for electron microscopy

PrgB₁₈₈₋₁₂₃₃ dimer fractions were thawed on ice and loaded on a Superdex 200 10/300 GL gel filtration column (GE Healthcare) equilibrated in 20 mM HEPES/NaOH pH 7.0 and 150 mM NaCl. Protein peak fractions, corresponding to the dimer, were diluted to 0.1-0.3 mg/mL and for the DNA-bound structures 120 bp ssDNA (Table S1) was added in 1: 1.2 molar ratio (protein:DNA) and samples were incubated for 15 minutes on ice. For both apo and DNA-bound samples, 4 µl of sample was applied to glow discharged Quantifoil 300 mesh 1.2/1.3 (Quantifoil) grids at 4 °C and 90-100% humidity, blotted for 1 s with blot force -5, and plunge-frozen in liquid ethane using a Vitrobot Mark IV (Thermo Fisher Scientific).

Cryo-EM data collection

Cryo-EM data were collected on an FEI Titan Krios transmission electron microscope (Thermo Fisher Scientific), operated at 300 keV that was equipped with a K2 direct electron detector. Data was collected by the AFIS method using the EPU software V2.8.0 (Thermo scientific) at a nominal magnification of 165,000x (0.82 Å pixel size). Data collection parameters are listed in Table S2. A total number of 2787 movie stacks were collected for apo PrgB and 1670 for DNA-bound PrgB.

Cryo-EM data processing

Cryo-EM data of apo PrgB and DNA-bound PrgB were processed in the same way, but separately using cryoSPARC (v3.2.0-3.3.1)⁽⁴⁵⁾. Beam-induced motion was corrected using standard settings, where start frame 1 was excluded, followed by per-micrograph contrast transfer function (CTF) estimation. For apo PrgB a subset of 170,826 particles from 500 micrographs were extracted with a box size of 384 Å. They were then picked using the blob picking tool with a 100-300 Å particle diameter. Picked particles were subjected to consecutive rounds of 2D classifications. Template particles were selected from representative 2D classes, and the subsequent full dataset was picked using the template picker tool and the input of the subset processing. PrgB with DNA was directly picked using blob picker with a 100-300 Å particle diameter and standard settings and extracted with 384 pix. PrgB without and with DNA were then separately subjected to 2D classifications resulting in a final number of 283,630 and 163,566 particles respectively. Particles from selected classes were combined and used in ab-initio reconstruction. The initial volume was then subjected to homogenous 3D refinement and the resolution was calculated using the gold standard Fourier shell correlation (FSC threshold, 0.143) and found to be 8 Å and 11 Å for the apo and DNA-bound structure, respectively. The volumes of apo and DNA-bound PrgB have been deposited in the Electron Microscopy Data Bank (EMDB Codes: EMD-16001 and EMD-16002).

The adhesion domain (PDB Code: 6EVU)⁽³²⁾ and stalk domain (PDB Code: 8BEG) were initially docked into the EM volume using Chimera⁽⁴⁶⁾ and subsequently run through Namdinator⁽⁴⁷⁾ using 10 Å resolution and standard settings. The output was then fitted in the EM volume in Chimera (v 1.15rc⁽⁴⁶⁾), where figures also were generated.

Western blotting

Cell pellets of each bacterial strain were harvested after induction and incubation overnight, and treated with lysozyme buffer (10 mM Tris, pH 8.0, 1 mM EDTA, 25% sucrose, 15 mg/ml lysozyme) for 30 min at 37°C. The lysozyme-treated bacterial lysates were then mixed with protein loading dye and boiled at 100°C for 12 min. The prepared samples were run on 8% SDS-PAGE, transferred to Western blot and probed with the PrgB antibody produced in rabbit.

Aggregation assay

E. faecalis strains were inoculated in BHI medium with the indicated antibiotics and cultured O/N at 37°C. Overnight cultures were diluted in a 1:100 ratio in BHI with the indicated antibiotics, 10 ng/mL cCF10, and 50 ng/mL nisin and dispensed into polystyrene cuvettes (Sarstedt) in 0.9 mL triplicates. These were incubated for 24 h at 37°C without agitation. Afterwards, the optical density of each sample was determined at 600 nm both before (OD_{sup}) and after (OD_{mix}) vigorously mixing of the bacterial culture by pipetting. The autoaggregation percentage was then calculated as follows: $100 \times [1 - (OD_{sup}/OD_{mix})]$ ^{(17),(29)}.

Biofilm assay

E. faecalis strains were inoculated in BHI with the indicated antibiotics and kept 16 h at 37°C. The next morning, they were diluted in a 1:100 ratio in BHI with the indicated antibiotics, 10 ng/mL cCF10, and 50 ng/mL nisin. 200 µl fractions were dispensed into a 96-well micro-titer plate (Costar) with 8 replicates per strain. 200 µL BHI fractions were used as blanks. The 96-well plate was then incubated aerobically at 37°C without agitation in a humidified chamber

for 24 h. The suspension was transferred to another 96-well plate to determine the optical density at 600 nm (OD_{600}). The plate containing the biofilm was washed with distilled water three times and then left to air dry at room temp for 2.5 h. The biofilm was stained with 100 μ l 0.1% (w/v) safranin (Sigma) at room temp for 20 min, then washed three times with distilled water and left to air dry at room temperature. Afterwards the absorbance was determined using a plate reader (BMG Labtech) at 450 nm. Biofilm production was calculated as an index of safranin staining of the cell biomass divided by absorbance of its optical density (OD_{450}/OD_{600})⁽⁴⁸⁾.

Conjugation assay

Donor (OG1RF) and recipient (OG1ES) strains were inoculated in BHI with the indicated antibiotics and incubated overnight at 37°C with agitation. Overnight cultured strains were refreshed in BHI without antibiotics in a 1:10 ratio, and donor strains were induced with 50 ng/mL nisin (Sigma). All strains were then incubated at 37°C for 1h without agitation. Afterwards each of the donor strains was mixed with the recipient cells in ratio of 1:10 and mated at 37°C statically for 3.5 h. These mixtures were then serially diluted with BHI and plated out in triplicates on BHI agar plates supplemented with tetracycline and spectinomycin (to select for donor cells), or with tetracycline, erythromycin, and streptomycin (to select for transconjugants). Plates were incubated at 37°C for 48 h, counted and enumerated for colony forming units (CFU). The plasmid transfer rate was determined as CFU of transconjugant over CFU of donor (Tc's/Donors)⁽³²⁾.

Electrophoretic Mobility Shift Assay (EMSA)

EMSA was carried out in the same way as previously described⁽³²⁾. 0.1 to 20 μ M PrgB (wild type and variants) were mixed with 50 nM 100 bp long double stranded DNA⁽³²⁾. Samples were incubated for 1 hour at 20 °C before loading them onto a 6% TBE-based native acrylamide gel for electrophoresis for 90 min at 50 V and 6 °C. Gels were subsequently stained with 3× GelRed (Biotium) in distilled water for 30 minutes and imaged with a Chemidoc system (BioRad). Quantification of the DNA bands after imaging was done in ImageLab (BioRad).

Statistical analysis

All shown *in vivo* data is from three independent experiments and was plotted and analyzed using GraphPad Prism (version 5.0) (GraphPad Software). The indicated error is the standard deviation over 3 biologically independent replicates.

Results

PrgB₅₈₄₋₁₂₃₃ contains 4 immunoglobulin-like domains

Previous bioinformatics and structural analysis of PrgB proposed that PrgB consisted of 3 domains; the previously crystallized polymer adhesin domain responsible for eDNA and LTA binding, and two domains with RGD (Arg-Gly-Asp) motifs⁽³²⁾. However, when we reanalyzed the PrgB sequence with the new structure-prediction tools that are available, such as AlphaFold⁽⁴⁹⁾, it became clear that this model was partially incorrect. PrgB seems to consist of an N-terminal disordered region (residues 35-197), followed by a newly identified coiled-coil (COI) domain (residues 198-257), the previously crystallized polymer adhesin domain (residues 261-558), a linker region (residues 559-582), 4 immunoglobulin (Ig)-like domains

(residues 583-1232) and finally the C-terminal disordered region containing the LPXTG motif (residues 1263-1305) that gets anchored to the cell wall (Fig. 1A). The Ig-like domains seem to come in pairs of two slightly different structures, denoted as CSA1-CSC1 (first pair) and CSA2-CSC2 (second pair), as named in InterPro (CSA from IPR026345; adhesin isopeptide-forming adherence domain, and CSC from IPR032300; cell-surface antigen C-terminal). To verify this updated domain organization of PrgB, we set out to experimentally determine its structure using a combination of X-ray crystallography and cryo-EM methods.

As described in Schmitt *et al.*⁽³²⁾, we were not able to produce full-length PrgB in *E. coli*, but instead produced and purified PrgB₁₈₈₋₁₂₃₃. This version of the protein only lacks the disordered N-terminal region and the LPXTG anchor and elutes from size exclusion chromatography in two peaks corresponding to dimeric and a monomeric PrgB⁽⁷⁾. The monomeric fraction was successfully used for crystallization trials. Crystals belonging to space group P2₁2₁2₁ appeared after 8-12 weeks, diffracted to 1.85 Å and contained 2 molecules in the asymmetric unit. The crystallographic phase problem was solved using molecular replacement with SspB (PDB: 2WOY) as a search model. Surprisingly, the resulting electron density lacked the previously crystallized polymer adhesin domain of PrgB, and instead only contained residues 584-1233. Likely, the polymer adhesin domain had been cleaved off in the crystallization drop before the crystals were formed. The modelled protein indeed consists of four Immunoglobulin (Ig)-like domains: two CSA and two CSC domains (Fig. 2A). Previous bioinformatics analysis showed that various homologous adhesin proteins contain different numbers of Ig-like domains⁽¹⁶⁾, but a DALI⁽⁵⁰⁾ analysis of PrgB₅₈₄₋₁₂₃₃ showed that there is no previously solved structure in the PDB that contains four of these Ig-domains coupled together. There are, however, homologous structures available with either 2 or 3 Ig-domains. The closest structural homologs are the C-terminal parts of Antigen I/II proteins from oral Streptococci, e.g. the surface protein AspA from *Streptococcus pyogenes*⁽⁵¹⁾ (PDB code: 4OFQ), which has 3 Ig-domains and an r.m.s.d. to the Ig-like domains from PrgB of 3.2 Å over 337 residues or the BspA protein from *Streptococcus agalactiae*⁽⁵²⁾ (PDB code: 4ZLP) which has 2 Ig-domains and an r.m.s.d. of 3.3 Å over 334 residues (see Table S3 for an overview of the highest ranked DALI hits). The r.m.s.d. decreases to ca. 1 Å if the individual Ig-like domains are superimposed upon each other.

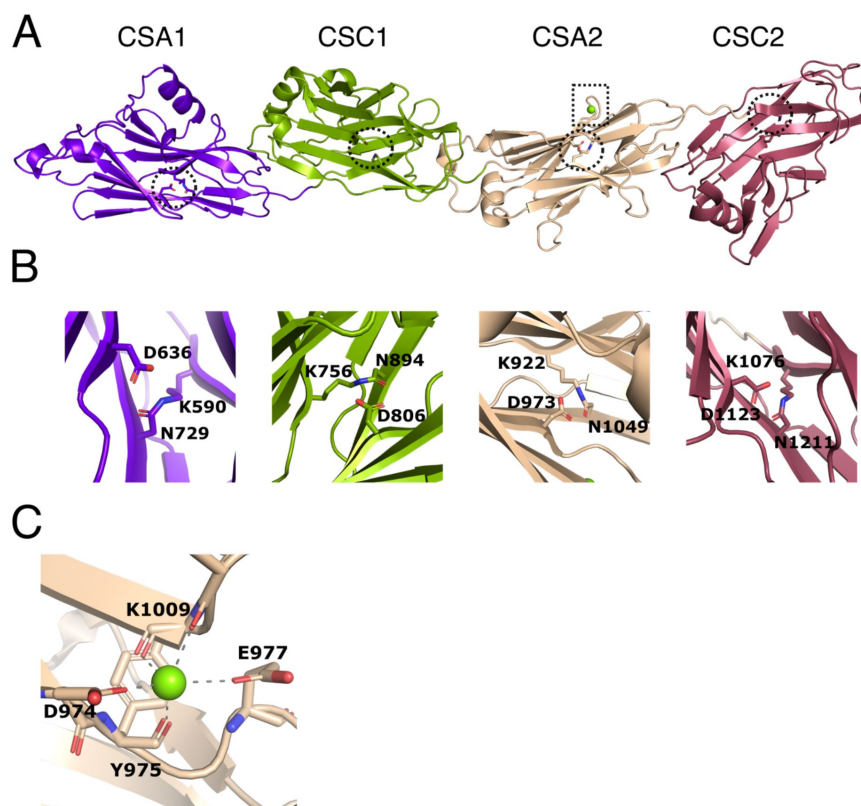


Figure 2.

Structure of Ig-like domains of PrgB. A) The Ig-like domains are arranged as tandem pairs (CSA1-CSC1 and CSA2-CSC2). B) Each Ig-like domain has an internal isopeptide bond (indicated by the striped circle in panel A) strengthening the structural integrity of the domain. Each of the four isopeptide bonds is between a lysine and an asparagine and further stabilized by an aspartic acid residue (highlighted residues are shown in stick representation). C) A conserved metal binding site is situated in the CSA2 domain (highlighted by a striped box in panel A), here modelled with a Mg^{2+} (green sphere).

In each of the Ig-like domains of PrgB, isopeptide bonds are formed between a lysine and an asparagine, a bond which is further stabilized by an aspartic acid (Fig. 2B). This feature is also present in various homologous Ig-like domains from Antigen I/II proteins^{(40),(51),(53)}. There is density in the conserved metal binding site of the CSA2 domain that has been suggested to bind Ca^{2+} in the Antigen I/II proteins. Refinement of our structure indicated that Mg^{2+} was the best fit to the density (Fig. 2C). The homologous C2 domains from AspA⁽⁵¹⁾, Pas⁽⁵⁴⁾, SpaP⁽⁵³⁾ and SspB⁽⁴⁰⁾, each have an extra structural feature termed the BAR (SspBadherence region) domain, which mediates adherence in these proteins. This BAR domain is absent in PrgB (Fig. S1).

DNA binding leads to large displacement of the polymer adhesin domain of PrgB

As we didn't manage to crystallize PrgB₁₈₈₋₁₂₃₃ without the loss of the polymer adhesin domain, we determined its structure via cryo-EM and single particle analysis. To determine the conformational changes of PrgB upon DNA binding, we collected two datasets: one without and one with ssDNA (120 bp). This yielded volumes with an average resolution of 8 and 11 Å, respectively. Even though dimeric fractions of PrgB were used for the grid preparations, the resulting maps of PrgB are monomeric, indicating that the dimer fell apart. Despite the relatively low resolution of the EM volume, mainly due to a preferred orientation of the protein on the cryo-EM grids, we could dock in the X-ray structures: the previously solved polymer adhesin domain (PDB code: 6EVU)⁽³²⁾ and the newly solved structure of PrgB₅₈₄₋₁₂₃₃. The resulting model of apo-PrgB shows that the polymer adhesin domain is folded back on top of the Ig-like domains (Fig. 3A). In the DNA-bound PrgB model the polymer adhesin domain has undergone a large displacement away from the Ig-like domains. There is also a larger volume for this domain, indicating that DNA has indeed been bound (Fig. 3B).

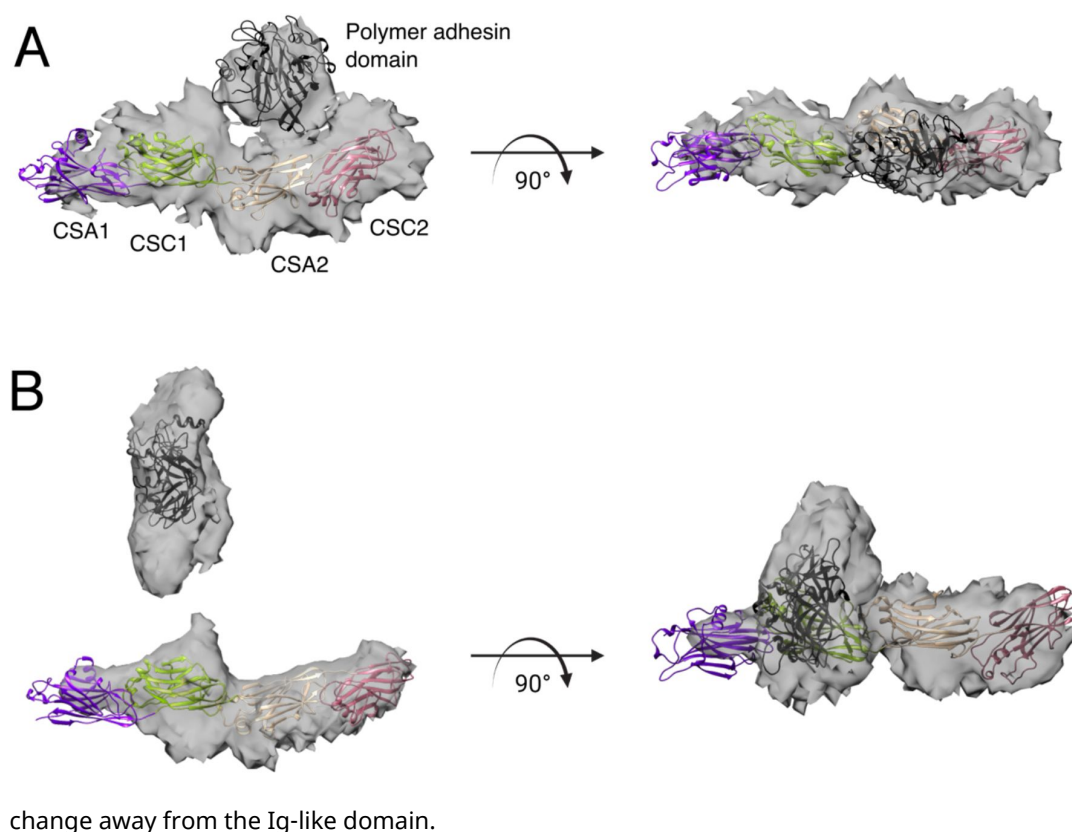


Figure 3.

Crystal structures of the PrgB polymer adhesin domain and Ig-like domains in cartoon representation docked into the volumes acquired by single particle cryo-EM. A) The PrgB₁₈₈₋₁₂₃₃ model shows the polymer adhesin domain folded back onto the Ig-like domains. B) In the DNA-bound PrgB₁₈₈₋₁₂₃₃ model the polymer adhesin domain has undergone a large conformation

In vivo assays

Based on the new structural insights for PrgB, we decided to study the importance of the newly defined coiled-coil domain and the Ig-like domains. This was done by complementing *E. faecalis* OG1RF pCF10ΔprgB with different prgB mutants (from a nisin-inducible plasmid) and characterizing their phenotypes in cellular aggregation, biofilm formation, and plasmid transfer efficiency. In line with previous experiments⁽¹⁷⁾, complementing pCF10ΔprgB with exogenous PrgB from the pMSP3545S vector rescues all phenotypes. Aggregation and biofilm formation are even slightly increased as compared to wild-type pCF10 (Fig. 4A-C, column 1 and 3), possibly due to a slightly increased production of PrgB (Fig. S2, lane 1 and 3).

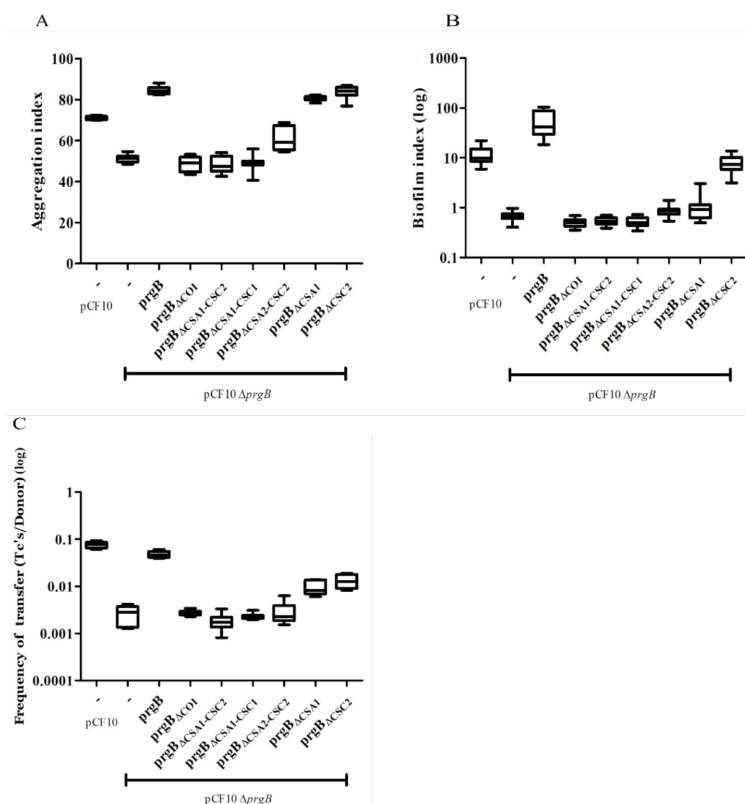


Figure 4.

In vivo phenotypes of PrgB variants. PrgB variants are expressed from the pMSP3545S-MCS vector in the OG1RF pCF10 Δ prgB background for phenotypic analysis with three assays: A) cellular aggregation B) biofilm formation and C) conjugation assays. For all assays OG1RF pCF10 carrying an empty vector serves as positive control and OG1RF pCF10 Δ prgB with an empty vector as negative control. The value of each column represents the average of three independent experiments and the error bars represent the standard error of the mean (SEM).

We found that PrgB without the newly identified coiled-coil domain could not rescue the aggregation phenotype of the deletion strain (Fig. 4A, column 4). Western blot analysis showed reduced levels of PrgB Δ COI as compared to wild type (Fig. S2) probably due to a decreased stability, which could explain the observed decrease in aggregation. Deletion of either the CSA1 or the CSC2 domain did not affect PrgB-mediated aggregation (Fig. 4A, column 8 and 9). However, complementation with PrgB Δ CSA2-CSC2 could only partially rescue the aggregation phenotype of the *E. faecalis* OG1RF pCF10 Δ prgB strain (Fig. 4A, column 7) and no rescue at all was seen in the PrgB variants with both the CSA1 and CSC1 domain deleted or without any Ig-like domains (CSA1-CSC1-CSA2-CSC2) (Fig. 4A, column 5, and 6).

As expected⁽¹⁷⁾, deletion of *prgB* also lead to a large decrease in biofilm formation (Fig. 4B). In line with our observations from the aggregation assays, PrgB with a deletion of either the coiled-coil domain or more than a single Ig-like domain could not rescue the *E. faecalis* OG1RF pCF10 Δ prgB biofilm formation phenotype. Only exogenous expression of PrgB Δ CSC2 can rescue the level of biofilm formation, but only to the level found in OG1RF pCF10, not to the level of exogenously expressed wild-type *prgB* (Fig 4B, column 1, 3 and 9). Intriguingly, the expression of exogenous PrgB Δ CSA1 in the OG1RF pCF 10 Δ prgB background did not restore biofilm formation, while it did restore the aggregation phenotype (Fig. 4A and 4B, column 8). In the conjugation assay, exogenous expression of the PrgB variants that failed to rescue the aggregation phenotype of the OG1RF pCF10 Δ prgB strain (PrgB Δ COI, PrgB Δ CSA1-CSC2, PrgB Δ CSA1-CSC1 and PrgB Δ CSA2-CSC2) were also found to have a decreased plasmid transfer efficiency (Fig. 4C, column 4-7). However, PrgB Δ CSA1 and PrgB Δ CSC2 could only partially rescue the conjugation rate of the OG1RF pCF10 Δ prgB strain (Fig. 4C, column 8-9), while they did rescue the aggregation phenotype.

The conserved binding cleft in the polymer adhesin domain is essential for conjugation and biofilm formation, but not for aggregation

To investigate the role of the binding cleft in the polymer adhesin domain, we introduced single, double, and triple mutations to alter its conserved residues. The resulting *prgB* mutants were exogenously expressed in the background of pCF10Δ*prgB* for functional complementation; or in the background of wild-type pCF10 to test for any potential dominant negative effects as previously observed for PrgB_{Δ246-558} (deletion of the polymer adhesin domain)⁽³²⁾. Notably, all the resulting *prgB* mutants fully restored the aggregation phenotype of pCF10Δ*prgB* (Fig. 5A), but did not rescue the defective biofilm formation, nor the reduced conjugation efficiency (Fig. 5B and C, column 2-6). No dominant negative effects on aggregation or biofilm formation could be observed when these variants were expressed in the wild-type pCF10 background (Fig. 5A and B, column 7-10), although slightly reduced conjugation rates were observed (Fig. 5C, column 7-10). We have previously shown that the polymer adhesin domain binds eDNA and that this binding correlates with both biofilm formation and conjugation efficiency. Therefore, we wanted to test whether eDNA binding was affected in these PrgB variants. To do so, we purified both wild-type PrgB₁₈₈₋₁₂₃₅ and the double mutant PrgB₁₈₈₋₁₂₃₃:S442A,N444A to compare their DNA binding affinities via electrophoretic mobility shift assays (EMSA). The results indicate that the introduced changes in PrgB did not affect its ability to bind eDNA, as the affinity for PrgB₁₈₈₋₁₂₃₃ and PrgB₁₈₈₋₁₂₃₃:S442A,N444A were the same within experimental error (Fig. S3).

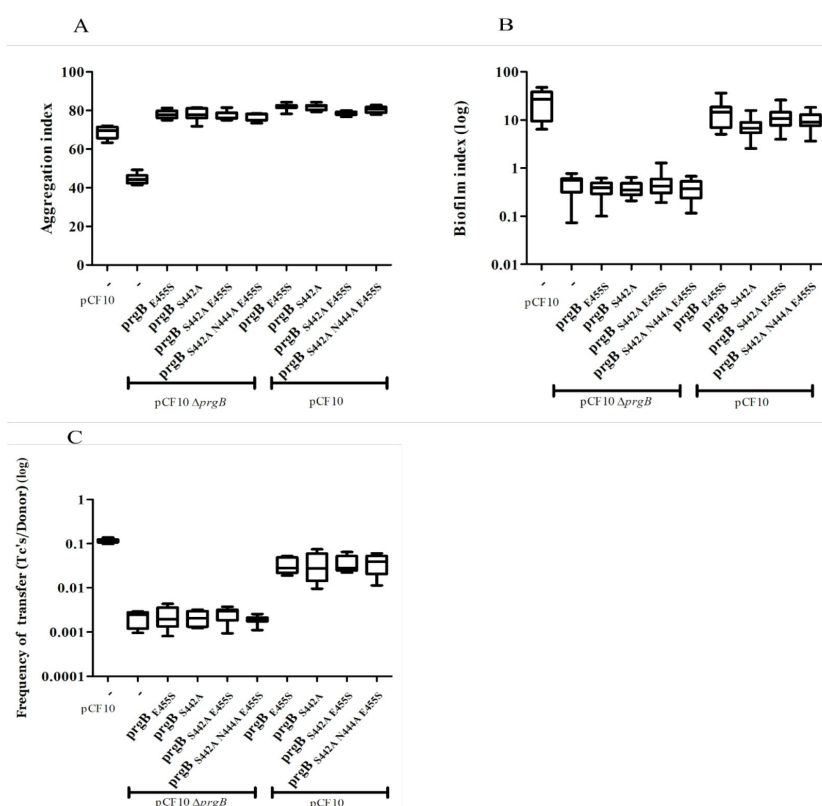


Figure 5.

In vivo phenotypes of PrgB variants with point mutation(s) in the conserved site of the polymer adhesin domain. PrgB variants are expressed from the pMSP3545S-MCS vector in the OG1RF pCF10Δ*prgB* or OG1RF pCF10 strain and analyzed in: A) cellular aggregation, B) biofilm formation, and C) conjugation assays. OG1RF pCF10 carrying the empty vector serves as positive control and OG1RF pCF10Δ*prgB* with the empty vector as negative control. The height of each column represents the average of three independent experiments and the error bars indicate the standard error of the mean (SEM).

Discussion

The presented data provides important insights for a widespread virulence factor in Gram-positive bacteria, since genes encoding PrgB homologs exist on a large number of

conjugative plasmids⁽¹⁶⁾. In this study, we expand our structural knowledge of PrgB beyond the polymer adhesin domain, to now encompass almost the entire protein.

The crystal structure of PrgB₅₈₃₋₁₂₃₃ shows that this part of PrgB consists of four tandemly arranged immunoglobulin (Ig)-like domains. These Ig-like domains show a high degree of structural homology to Streptococcal surface proteins, usually found in the oral cavity⁽¹⁶⁾.

These homologous proteins have been indicated to bind various molecules, such as fimbria, collagen and salivary agglutinin (SAG, also designated as glycoprotein 340), and these binding interactions have been shown to be vital for their function^{(40),(54)–(57)}. However, we have not found any evidence that the Ig-like domains of PrgB bind to a specific substrate in our own experiments, nor have we found this in other reports. PrgB also does not contain a BAR domain, which is crucial for stable interactions between e.g. SspB from *S. gordonii* and Mfa-1 of *P. gingivalis*^{(58),(59)}. However, since Ig-like domains are known to bind a large variety of ligands⁽⁶⁰⁾, we don't exclude the possibility that ligands for the Ig-like domains in PrgB will be found in the future. However, the only ligands that have been verified to interact with PrgB so far are eDNA and LTA, which have high affinity to the polymer adhesin domain⁽³²⁾.

The crystal structure of the Ig-like domains from PrgB, PrgB₅₈₃₋₁₂₃₃, was complemented by single particle analysis of PrgB₁₈₈₋₁₂₃₃ via cryo-EM (Fig. 3). Despite the low resolution of the EM volumes, we could dock in the high-resolution crystal structures of the Ig-like domains and the polymer adhesin domain. eDNA binding is a key step in PrgB mediated cellular aggregation and biofilm formation⁽³²⁾, and the models obtained from our cryo-EM volumes provided the first mechanistical insight into the conformational changes that occur during this step. The polymer adhesin domain is nestled to the side of the Ig-like domains (Fig. 3A), likely reflecting how PrgB is present on the outside of the cell wall in the absence of a binding partner. Possibly, the folding back of the polymer adhesin domain is protecting it from proteolytic cleavage (see below). Our model of PrgB bound to DNA suggests that the polymer adhesin domain gets displaced from the Ig-like domains upon eDNA binding (Fig. 3B). This would expose the positively charged surface of the polymer adhesin domain and allow it to wrap the eDNA around itself to condense it, as we observed in previous experiments^{(32),(33)}. We speculate that this in turn could also expose the PrgA-binding site, predicted to be located in the flexible region between the polymer adhesin domain and the CSA1 domain⁽¹⁵⁾. This would allow PrgA-mediated cleavage and removal of the polymer adhesin domain⁽¹⁵⁾. How this cleavage is regulated is not yet understood, but the folding back of the polymer adhesin domain onto the Ig-like domains in the absence of substrate could be part of this regulation.

We have now obtained a structural basis to interpret almost all phenotypic data that is available for PrgB. Unfortunately, most of the mutants that were previously described did not correlate well with the newly determined domain boundaries. Therefore, we decided to create specific deletion mutants that were based on the new PrgB structure to determine the role of the various domains. At the N-terminus of PrgB, before the polymer adhesin domain, there is a predicted coiled-coil region (Fig. S4A) that we wanted to investigate. To our surprise, the expression of PrgB_{ΔCOI} could not rescue any of the aggregation, biofilm formation or conjugation phenotypes from a *prgB* deletion strain. However, in our experiments PrgB_{ΔCOI} was only produced at low levels (Fig. S2), indicating that the coiled-coil region might be important for protein production and/or stability. Deletion of all Ig-like domains (PrgB_{ΔCSA1-ΔCSC2}) renders the protein incapable to support aggregation, biofilm formation and conjugation. However, exogenous expression of *prgB* with single Ig-like domain deletions, *prgB*_{ΔCSA1} and *prgB*_{ΔCSC2} in the pCF10Δ*prgB* background, restore cellular aggregation, while they do not rescue biofilm formation and conjugation (Fig. 4). This was unexpected, as the polymer adhesin domain is predicted to mediate all the functions that were tested in these assays: aggregation, biofilm formation and conjugation^{(17),(32)}. Our

results, however, indicate that it is important to have all Ig-like domains present and properly folded. The cellular aggregation assays seem to indicate that PrgB can function when at least 3 Ig-like domains are present, as expression of both *prgB* Δ CSA1 and *prgB* Δ CSC2 can complement pCF10 Δ *prgB*, but unfortunately this assay is not suitable to detect small changes. Based on the biofilm formation and conjugation efficiency assays, which are more sensitive, we therefore conclude that even removing a single Ig-like domain strongly decreases the function of PrgB.

The conserved Ser-Asn-Glu site in the negatively charged cleft of the polymer adhesin domain intrigued us, as its function is unknown. Any changes that we made in these conserved residues resulted in a PrgB variant that didn't facilitate biofilm formation or conjugation (Fig. 5). Surprisingly these PrgB variants did fully support cellular aggregation. This is thought-provoking, since various literature has shown that the PrgB functions in cellular aggregation, biofilm formation and conjugation are strongly correlated. However, even a single point mutation in this conserved site produced a PrgB variant that completely separates the cellular clumping phenotype from biofilm formation and conjugation. *In vitro* experiments showed that these point mutations did not impair PrgB binding to eDNA (Fig S3). A similar phenotypic pattern was observed with PrgB Δ CSA1 and PrgB Δ CSC2 which both could fully rescue aggregation but not biofilm formation. Our data therefore strongly indicates that PrgB has additional role(s), besides mediating cellular aggregation, to further support conjugation and biofilm formation. We therefore propose that PrgB performs at least one, currently unknown, additional function besides binding to eDNA and/or LTA from the cell wall of the recipient cell. It is highly likely that at least one of these additional functions is mediated by the conserved site in the polymer adhesin domain.

As described in the introduction, there is a plethora of *prgB* mutants made (see Fig. S4) and phenotypically analyzed, predominantly by the group of Prof. Gary Dunny. In supplementary table S4 we provide a summary of all *prgB* mutants that we have identified in the literature and our brief reinterpretation based on our new structural knowledge. Below, we will discuss a selected number of these mutants in detail.

As expected, most mutants that have an insertion or a mutation in the polymer adhesin domain (Fig. S4A) show a loss of protein function. Shortly, insertions at amino acids (a.a.) 358 and 359 are on the surface of the protein and likely leads to steric clashes that prevent the domain from binding to eDNA and LTA, whereas insertions at a.a. 439, 473, 517, and 546 are all in secondary structures that are central parts of the polymer adhesin domain and therefore are likely to disrupt folding of this domain.

The RGD motifs in PrgB were previously proposed to be involved in integrin binding and to promote adherence to human neutrophils, as well as internalization in cultured intestinal epithelial cells^{(25),(61)}. The new domain classification showed that these two RGD motifs are in CSA1 and CSA2, respectively (Fig. 1A). The structure that we determined shows that these two sequence motifs are not surface exposed, but instead play an important role in the structural integrity of the interfaces between CSA1 and CSC1, and between CSA2 and CSC2 (Fig. 6). Mutations in these motifs would thus very likely destabilize the folding of the tandem Ig-domains, which would explain the decreased PrgB biofilm formation observed in these strains^{(18),(23)}. Thus, our new data strongly argues against the previously proposed direct binding interaction between the RGD sequences and integrins of host origin.

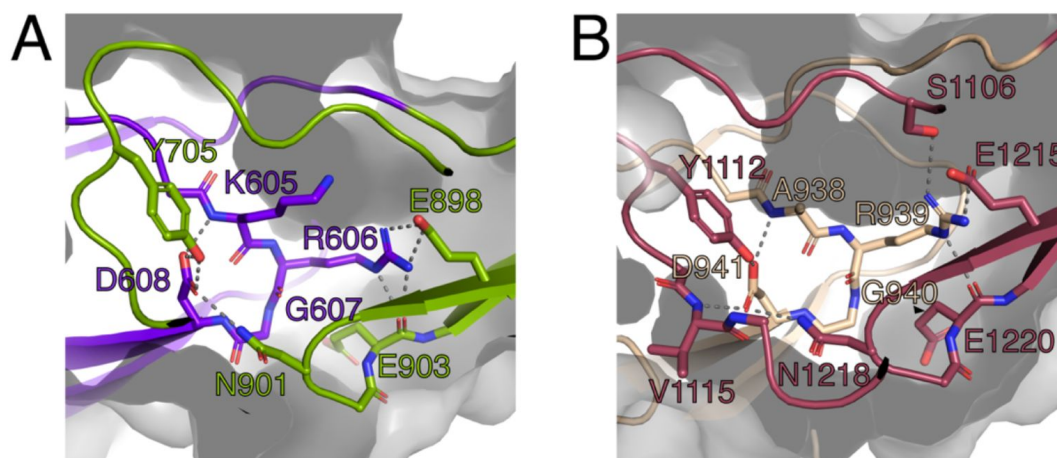


Figure 6.

The RGD motifs play a role in the structural integrity of PrgB. A) Close up view of the RGD motif in the CSA1 domain (purple). Most of the important interactions, mainly hydrogen bonds, formed by this motif is to residues on the CSC1 domain (green). B) Close up view of the RGD motif in the CSA2 domain (sand colored). Most of the important interactions, mainly hydrogen bonds, formed by this motif is to residues on the CSC2 domain (dark red). Potential hydrogen bonds are marked by lines and the surface of the protein indicated by transparent gray.

Deletion of residues 993-1138 leads to a loss of about half of both the CSA2 and CSC2 domains. Therefore, we were surprised to find that this mutant was described to behave like wild type in aggregation assays⁽³¹⁾. Possibly the remaining parts of CSA2 and CSC2 form a (unfolded) linker region of similar length to the tandem CSA2-CSC2 structure, which allows PrgB to retain its function of promoting aggregation. Similarly, PrgB_{Δ668-1138}, corresponding to a complete removal of CSC1 and CSA2 and approximately half of both CSA1 and CSC2 domains, could still support PrgB mediated aggregation. Potentially the remaining residues (ca 180 amino acids) could also form an unfolded linker region allowing for the variant to retain its aggregation phenotype. For PrgB_{Δ993-1138} and PrgB_{Δ668-1138}, unfortunately no biofilm formation or conjugation efficiency were reported, but based on our results it is likely that those capabilities would have been severely compromised.

Taking past and present results into account combined with our new structural insights, we propose a new mechanistic model for the function of PrgB. Our EM data indicates that the Ig-domains provide structural support for the polymer adhesin domain in the absence of bound ligand(s) and our *in vivo* data show that removal of one of the Ig-like domains does not largely affect PrgB function. Previous data also indicates that parts of the Ig-like domains can be deleted without affecting the function of PrgB. We therefore hypothesize that the Ig-like domains provide two important features to the protein. First, a rigid stalk that is needed to present the polymer adhesin domain at the correct distance from the cell wall. Second, providing protection from PrgA-mediated cleavage when the polymer adhesin domain is folded on top of the Ig-like domains. Binding of eDNA (and likely also LTA) releases the polymer adhesin domain from this stalk domain, and with that likely also exposes the predicted PrgA proteolysis site in the linker region between the polymer adhesin domain and the CSA1 domain^{(15),(16)}. The exact distance of the polymer adhesin domain from the cell wall does not seem important for LTA-binding, since aggregation can still take place even with partially disrupted Ig-like domains. However, the length of the stalk may be very important when it comes to facilitating both biofilm formation and conjugation. This

indicates that effective mating-pair formation in conjugation may require a PrgB-mediated function distinct from pure aggregation, something that is further shown by the findings of the mutations in the conserved site in the polymer adhesin domain.

Besides providing a structural basis to explain about 30 years of work on PrgB, we here also uncovered that the conserved Ser-Asn-Glu site in the polymer adhesin domain likely provides additional functionality to PrgB that is needed for optimal biofilm formation and conjugation, but that does not affect cellular aggregation. To fully investigate the function of this conserved site in PrgB and other homologous virulence factors from Gram-positive bacteria, remains an exciting question to address in future research.

Table 1.

X-ray data collection and refinement statistics. Values within parenthesis correspond to the highest resolution shell.

Data collection summary	PrgB ₅₈₄₋₁₂₃₃
Space group	P2 ₁ 2 ₁ 2 ₁
Cell dimensions	
a, b, c (Å)	76.0, 87.5, 224.8
α, β, γ (°)	90, 90, 90
Resolution (Å)	47.3 – 1.84 (1.91 – 1.84)
Completeness (%)	99.6 (99.5)
R _{meas} (%)	0.03 (0.99)
I/σ (I)	12.1 (1.0)
CC(1/2)*	1 (0.67)
Redundancy	2.0 (2.0)
No. unique reflections	130301 (12877)
Refinement summary	
Resolution (Å)	47.3 – 1.84
R _{work} (%)	20.9
R _{free} (%)	24.5
Number of atoms	
protein	10080
water	816
other ligands	2
B-factors	
protein	44.9
water	47.7
other ligands	48.3
r.m.s. deviations	
Bond lengths (Å)	0.006
Bond angles (°)	0.79
Ramachandran statistics	
outliers (%)	0
allowed (%)	1.5
favoured (%)	98.5

Supporting information

supplements/513195_file02.pdf Supplementary tables and
figuressupplements/513195_file03.xlsx Supplemental Table 4

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CRedit statement

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Author information

1. Wei-Sheng Sun

Department of Medical Biochemistry and Biophysics, Umeå University, SE-90187 Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden
ORCID iD: [0000-0001-9738-8862](https://orcid.org/0000-0001-9738-8862)

2. Lena Lassinantti

Department of Medical Biochemistry and Biophysics, Umeå University, SE-90187 Umeå, Sweden
ORCID iD: [0000-0001-5470-591X](https://orcid.org/0000-0001-5470-591X)

3. Michael Järvå

Department of Medical Biochemistry and Biophysics, Umeå University, SE-90187 Umeå, Sweden
ORCID iD: [0000-0002-8519-5622](https://orcid.org/0000-0002-8519-5622)

4. Andreas Schmitt

Department of Medical Biochemistry and Biophysics, Umeå University, SE-90187 Umeå, Sweden

5. Josy ter Beek

Department of Medical Biochemistry and Biophysics, Umeå University, SE-90187
Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå,
Sweden
ORCID iD: [0000-0003-4165-9277](https://orcid.org/0000-0003-4165-9277)

6. Ronnie P-A Berntsson

Department of Medical Biochemistry and Biophysics, Umeå University, SE-90187
Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå,
Sweden
ORCID iD: [0000-0001-6848-322X](https://orcid.org/0000-0001-6848-322X)

Editors

Reviewing Editor

Michael Eisen

University of California, Berkeley, United States of America

Senior Editor

Michael Eisen

University of California, Berkeley, United States of America

Reviewer #1 (Public Review):

Sun and colleagues outline structural and mechanistic studies of the bacterial adhesin PrgB, an atypical microbial cell surface-anchored polypeptide that binds DNA. The manuscript includes a crystal structure of the Ig-like domains of PrgB, cryo-EM structures of the majority of the intact polypeptide in DNA-bound and free forms, and an assessment of the phenotypes of *E. faecalis* strains expressing various PrgB mutants.

Generally, the study has been conducted with a good level of rigor, and there is consistency in the findings. However, I do have some specific technical concerns relating to the study. Although the PX work has been expertly undertaken, the Cryo-EM structures reported are both at ~10-angstrom resolution. Visual inspection indicates that the positioning of the PrgB domains into the EM envelopes is somewhat questionable and this needs to be addressed. The narrative of the manuscript very much hinges on this being correct. In addition, wrt the PrgB mutant studies, it could be that the loss of function observed in specific mutants is simply a consequence of mutation-induced misfolding of those polypeptides. Experimental evidence supporting the direct interaction between the PAD and the stalk domains in PrgB is also lacking.

Reviewer #2 (Public Review):

Having previously solved the X-ray crystallographic structure of the polymer adhesin domain (PAD) of PrgB from *E. faecalis*, the authors looked to build on that work by crystallizing a nearly full-length construct of PrgB. Though they were successful in their crystallization endeavors, the crystal contained only what was previously thought to be two domains with RGD motifs. The authors' high-resolution structure shows that in fact the C-terminal portion of PrgB is made up of four immunoglobulin-like domains. The authors then set out to collect single-particle cryoEM data in a bid to obtain a full-length structure of PrgB,

both in the presence and absence of ssDNA. The authors were only able to obtain quite low-resolution data, which they fit their crystal structures into. The authors then used these structures to inform the design of novel deletion mutants and point mutations, as well as to rationalize years of phenotypic data from other published mutants.

The X-ray crystallographic structure is beautiful and in combination with their in vivo data allowed them to propose a model where PrgB positions cells at an appropriate distance for conjugation. The cryoEM data are not convincing in their current state, and I, therefore, don't believe that their model of the immunoglobulin domains acting to protect the PAD domain of PrgB from PrgA is well supported. Perhaps there are 2D classes or other data that make a case for the fit of the crystal structures into the cryoEM volumes, but without a PAD deletion or perhaps a dataset including a PAD-specific antibody, I don't feel the fit is supported.

The in vivo experiments appear to be done well and the authors' discovery that the Ser-Asn-Glu is not important for generalized aggregation but has an additional yet unknown role in conjugation and biofilm formation is exciting and well supported by their data.

Author Response:

First of all, we would like to thank the reviewers for their work. We appreciate the constructive review comments and useful suggestions to further improve our article.

The main criticism on our manuscript, from both reviewers, is that the cryo-EM structures are of low resolution and that the fit of the crystallographic structures of the PAD and the stalk domain into these low-resolution structures is questionable. We would like to point out that the cryo-EM data, and the conclusions from it, are not essential for the main conclusions of the article. All mutants that we made in this study were designed based on the structural data obtained from the high-resolution X-ray structures, with no input from the low-resolution cryo-EM docked models. We chose to include the cryo-EM data since it allowed us to speculate about the interaction between the PAD and the stalk domain of PrgB, domains that we have separately determined the structures of via X-ray crystallography. We agree with the reviewers that further experiments are needed to verify this potential interaction. Therefore, we will perform additional biochemical assays to investigate the proposed interaction. We will also try to optimize the cryo-EM data to hopefully allow for a more reliable fit of our high-resolution crystallographic structures. Once that is done, we will submit a revised version of the manuscript.

On behalf of all authors,

Ronnie Berntsson